

SIBI, Pakistan

WMO: 416970

Lat: 29.550N

Lon: 67.883E

Elev: 440

StdP: 14.46

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.7	45.2	19.4	14.8	66.5	24.7	19.1	63.7	11.1	64.8	8.8	61.6	0.1	N/A	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	20.4	115.8	82.9	113.2	82.8	110.9	82.1	93.2	108.3	91.0	106.8	88.9	104.6	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
90.1	221.4	106.1	87.5	203.4	104.2	85.3	189.1	101.7	61.6	109.1	58.2	107.0	55.3	104.4	100.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	7.1	5.4	DB	38.5	120.7	3.8	2.5	35.8	122.5	33.6	124.0	31.4	125.4	28.6	127.2	(4)
			WB	34.0	96.2	2.8	3.6	31.9	98.8	30.3	101.0	28.7	103.0	26.7	105.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	59.1	65.3	75.8	88.0	97.9	101.1	98.1	95.2	92.8	84.6	72.3	62.1	(6)
(7)		DBStd	15.33	4.04	5.38	6.72	6.42	5.16	4.39	4.22	3.55	3.45	4.69	4.80	4.41	(7)
(8)		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	365	187	56	4	0	0	0	0	0	0	0	2	116	(9)
(10)		CDD50	11959	284	427	799	1140	1486	1532	1491	1400	1283	1074	669	374	(10)
(11)		CDD65	6847	5	63	338	690	1021	1082	1026	935	833	609	221	24	(11)
(12)		CDH74	109084	76	564	3569	10215	17886	19531	17928	15553	13248	8061	2124	329	(12)
(13)		CDH80	76079	3	154	1666	6534	13456	15213	13472	11103	8982	4677	767	52	(13)
(14)	Wind	WSAvg	1.0	0.5	0.9	1.0	1.3	1.5	1.4	1.5	1.4	0.9	0.7	0.4	0.4	(14)
(15)	Precipitation	PrecAvg	4.8	0.3	0.3	0.5	0.4	0.0	0.4	1.4	0.8	0.4	0.0	0.0	0.1	(15)
(16)		PrecMax	13.3	2.2	2.1	3.6	2.2	0.1	3.0	4.3	3.5	2.8	0.4	0.3	0.8	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.0	0.5	0.5	0.9	0.6	0.0	0.8	1.5	1.2	0.8	0.1	0.1	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	89.2	102.7	112.2	119.7	120.2	113.6	108.7	107.9	104.1	93.5	84.9	(19)
(20)		0.4%	MCWB	58.7	62.5	74.6	74.0	85.1	83.6	86.5	87.4	80.7	76.3	66.9	65.1	(20)
(21)		2%	DB	75.8	84.6	97.2	108.9	115.9	116.2	111.1	106.2	105.2	101.6	89.9	80.9	(21)
(22)		2%	MCWB	57.9	63.9	72.1	74.5	83.3	82.6	85.9	85.4	80.1	76.1	69.3	61.2	(22)
(23)		5%	DB	73.3	80.1	92.6	105.4	113.3	113.9	109.1	104.7	103.4	99.1	87.2	77.7	(23)
(24)		5%	MCWB	57.1	61.0	69.8	73.7	81.4	83.7	85.4	84.8	80.4	75.7	68.5	59.7	(24)
(25)		10%	DB	70.4	76.8	88.6	102.2	110.6	111.6	106.8	103.1	101.6	96.5	84.7	74.7	(25)
(26)		10%	MCWB	55.7	59.9	67.5	73.5	79.9	83.2	85.1	84.5	80.5	73.7	66.8	59.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	69.2	77.9	85.8	97.0	95.3	94.7	93.1	88.3	84.2	76.6	68.0	(27)
(28)		0.4%	MCDWB	71.1	82.2	99.0	104.0	110.3	111.9	107.4	105.1	100.3	97.7	86.1	80.3	(28)
(29)		2%	WB	60.3	65.7	73.5	80.8	92.7	92.8	91.7	89.8	86.0	81.3	73.4	63.9	(29)
(30)		2%	MCDWB	71.9	79.2	93.7	100.5	108.0	109.8	106.0	102.6	98.7	95.8	85.8	77.2	(30)
(31)		5%	WB	58.6	63.8	71.2	77.8	89.2	90.8	89.5	87.9	84.6	79.6	71.0	62.0	(31)
(32)		5%	MCDWB	70.6	76.6	89.4	98.5	107.1	108.4	104.1	100.4	97.7	93.7	83.6	74.8	(32)
(33)		10%	WB	57.1	61.8	69.2	75.3	85.8	88.6	87.6	86.3	83.5	77.7	68.5	60.0	(33)
(34)		10%	MCDWB	68.4	73.6	85.2	96.2	105.5	106.1	102.4	98.8	96.7	90.3	80.7	72.6	(34)
(35)	Mean Daily Temperature Range		MDBR	23.3	21.6	20.8	22.8	23.4	20.4	17.2	16.1	18.1	23.1	24.7	25.5	(35)
(36)		5% DB	MCDBR	26.9	26.3	25.1	28.0	26.3	23.8	18.7	17.2	19.6	24.2	25.9	29.2	(36)
(37)			MCWBR	13.7	12.2	11.3	12.2	13.0	11.1	7.8	6.6	6.8	9.6	12.6	15.1	(37)
(38)			MCDBR	23.7	22.9	23.6	24.1	23.8	21.2	17.8	16.5	17.1	20.9	23.0	27.7	(38)
(39)		5% WB	MCWBR	13.6	11.6	11.8	13.6	16.5	12.9	9.2	8.4	7.9	9.7	13.4	16.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.458	0.488	0.540	0.553	0.650	0.772	0.688	0.532	0.454	0.451	0.393	(40)	
(41)		taud	2.045	1.947	1.879	1.796	1.774	1.626	1.473	1.607	1.862	1.950	1.937	2.101	(41)	
(42)		Ebn at Noon	253	252	253	245	242	219	193	207	238	249	236	252	(42)	
(43)		Edh at Noon	45	54	62	69	71	82	96	83	62	53	50	41	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1203	1424	1826	2142	2382	2358	2102	2014	1953	1725	1342	1164	(44)	
(45)		RadStd	72	144	117	100	77	97	94	99	62	54	72	65	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page